

Design and Performance Evaluation of a Hybrid Free Space Optics and Fiber Communication System with Adaptive Switching.

Abstract:-

Free Space Optical (FSO) communication offers high data rates and cost-effective deployment; however, its performance is significantly affected by atmospheric conditions such as fog, rain, and turbulence. To overcome these limitations and enhance link reliability, this paper proposes a **hybrid FSO–Fiber communication system with an automatic switching mechanism**. The system prioritizes the FSO link under normal conditions and dynamically switches to a Fiber optic link when the FSO signal quality degrades.

The proposed model is designed and simulated using **OptiSystem**, incorporating key components such as optical transmitters, FSO channel, optical Fiber link, photodetector, low-pass filter, and a control unit for link selection. A feedback-based decision mechanism is implemented by monitoring the received signal quality and generating a binary control signal to operate an optical switch. The system ensures that only one communication path is active at a time, thereby improving efficiency and avoiding unnecessary power consumption.

Performance evaluation is carried out under varying atmospheric attenuation conditions to emulate real-world FSO impairments. The results demonstrate that the hybrid system significantly improves link availability and reliability compared to standalone FSO systems. The automatic switching mechanism effectively maintains continuous data transmission by seamlessly transferring communication from the FSO link to the Fiber link during adverse conditions.

This work highlights the potential of hybrid optical communication systems for robust and high-speed data transmission, making them suitable for next-generation communication networks.

Introduction:-

The rapid growth in data traffic and the increasing demand for high-speed communication systems have driven the need for advanced optical communication technologies. Free Space Optical (FSO) communication has emerged as a promising solution due to its ability to provide high bandwidth, license-free spectrum, and ease of deployment. FSO systems transmit data through the atmosphere using optical signals, making them suitable for applications such as last-mile connectivity, disaster recovery, and temporary network setups. However, the performance of FSO links is highly sensitive to atmospheric conditions such as fog, rain, haze, and turbulence, which can cause severe attenuation and degrade link reliability.

On the other hand, optical Fiber communication offers highly reliable and stable transmission with minimal attenuation and interference. Despite its advantages, Fiber optic systems involve high installation costs and lack flexibility in deployment, especially in challenging terrains or urban environments where laying Fiber is difficult. Therefore, relying solely on either FSO or Fiber communication presents limitations in achieving both reliability and flexibility.

To address these challenges, hybrid communication systems combining FSO and optical Fiber links have been proposed. Such systems aim to leverage the advantages of both technologies—utilizing the high-speed and flexible nature of FSO under favourable conditions, while switching to the more reliable Fiber link

during adverse atmospheric conditions. This hybrid approach significantly enhances system availability and ensures uninterrupted communication.

In this work, a hybrid FSO–Fiber communication system with an automatic switching mechanism is proposed and simulated using OptiSystem. The system prioritizes the FSO link during normal operation and dynamically switches to the Fiber link when the received signal quality falls below a predefined threshold. A feedback-based control mechanism is implemented using photodetection, filtering, and decision-making blocks to generate a binary control signal for the optical switch.

The proposed system is evaluated under varying atmospheric conditions to analyse its performance and reliability. The results demonstrate that the hybrid system effectively mitigates the limitations of standalone FSO communication and provides a robust solution for next-generation high-speed communication networks

1. Fiber optic link

Block Diagram :-

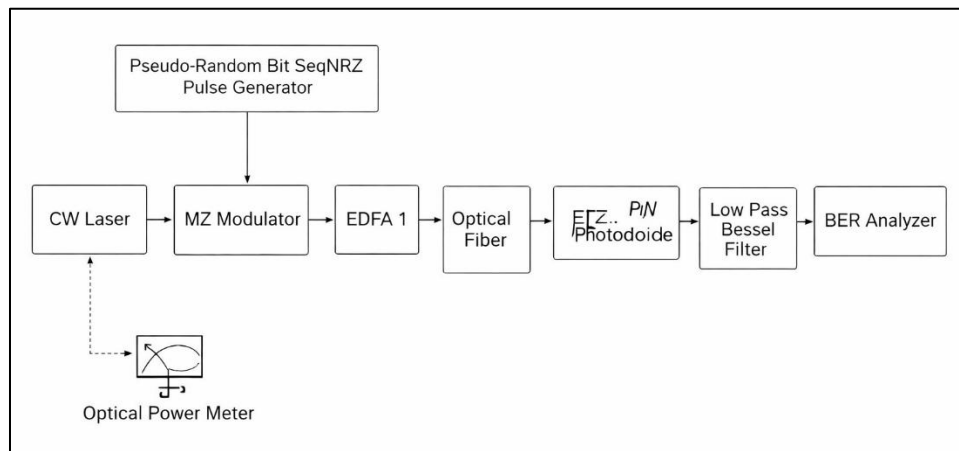


Figure 1 Block diagram of Fiber optic Communication

Simulation:-

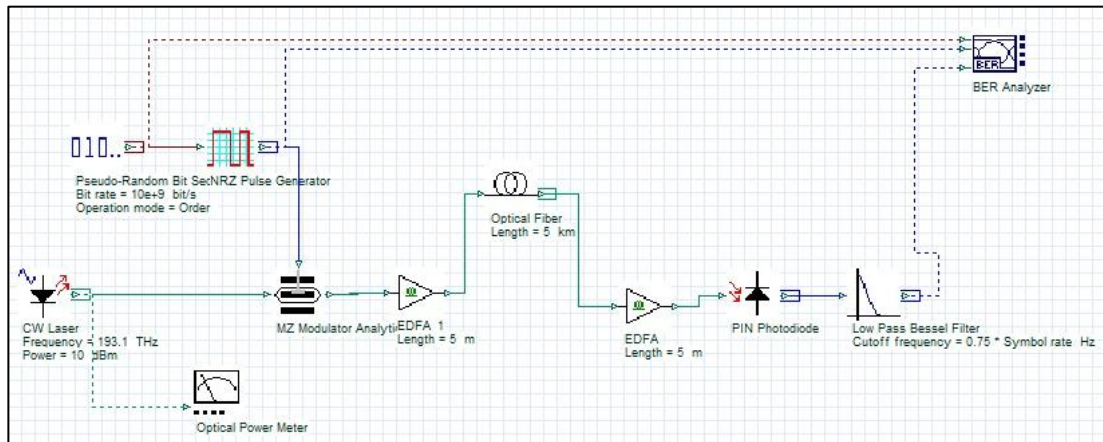


Figure 2 Circuit diagram of Fiber Optic Communication

Results:-

Parameters:-

Bit rate = 10 Gbit/s

CW frequency = 193.1 THz

CW Laser Power = 25 dBm

Symbol rate = 10 G Symbol/sec

Extinction ratio of MZ Modulator = 30 dB

EDFA length = 5 m

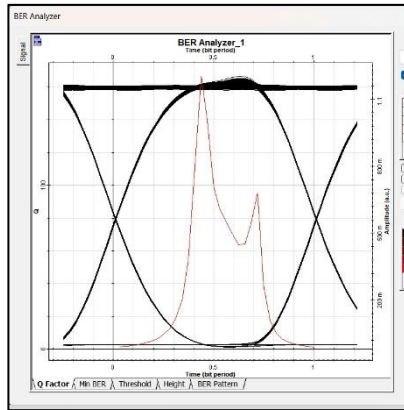


Figure 3 EYE diagram:- Range 2km

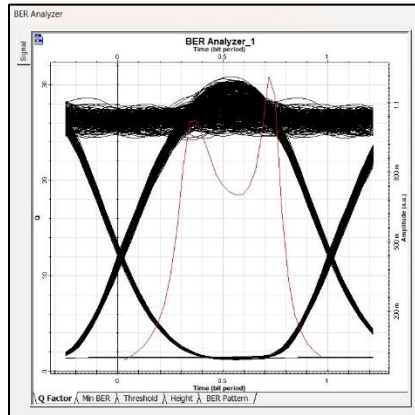


Figure 4 EYE Diagram:- Range 5km

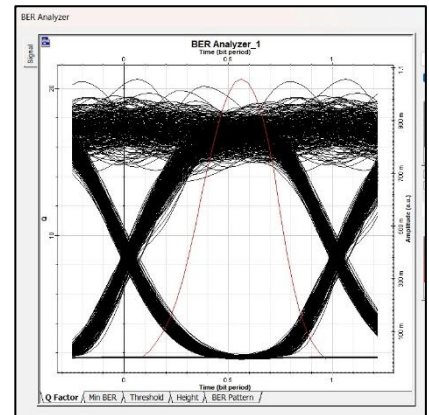


Figure 5 EYE Diagram:- Range 10km

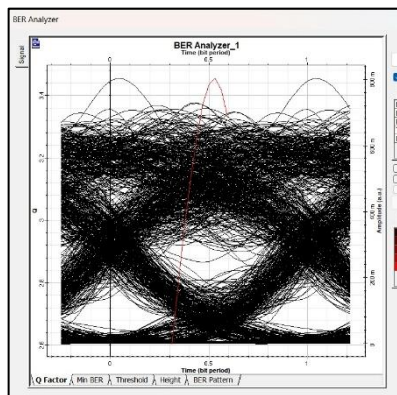


Figure 6 EYE diagram:- Range 20km

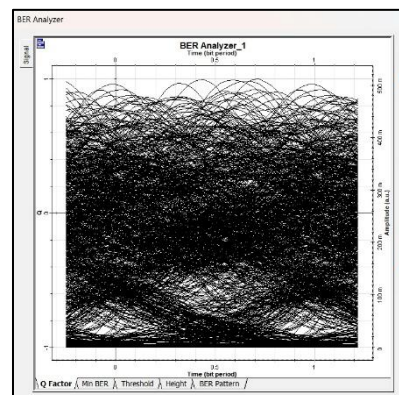


Figure 7 EYE diagram:- Range 30km

Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Threshold
0.2Db/Km	2	271.43	0	0.40607	0.044
	5	76.918	0	0.3653	0.031
	10	37.5327	1.15889e-308	0.30044	0.08578
	20	17.37	4.28374e-68	0.21808	0.0218
	30	5.63642	7.25109e-09	0.0935	0.057

2. FSO link

Block Diagram:-

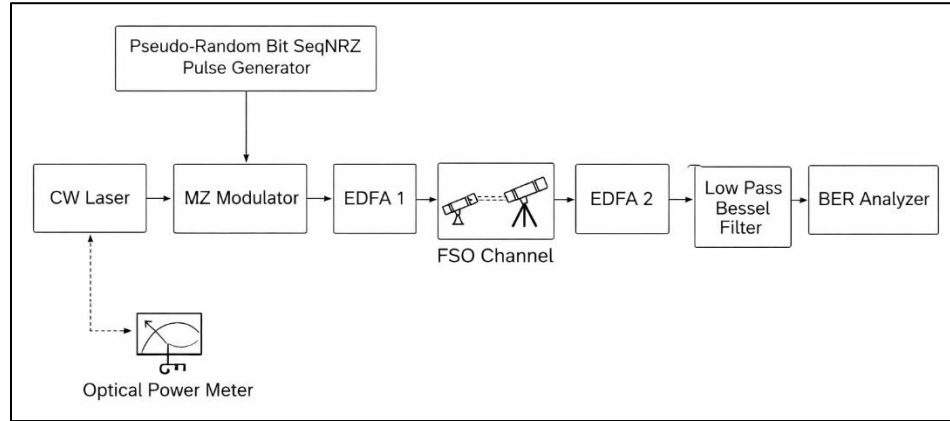


Figure 8 Block diagram of FSO

Simulation :-

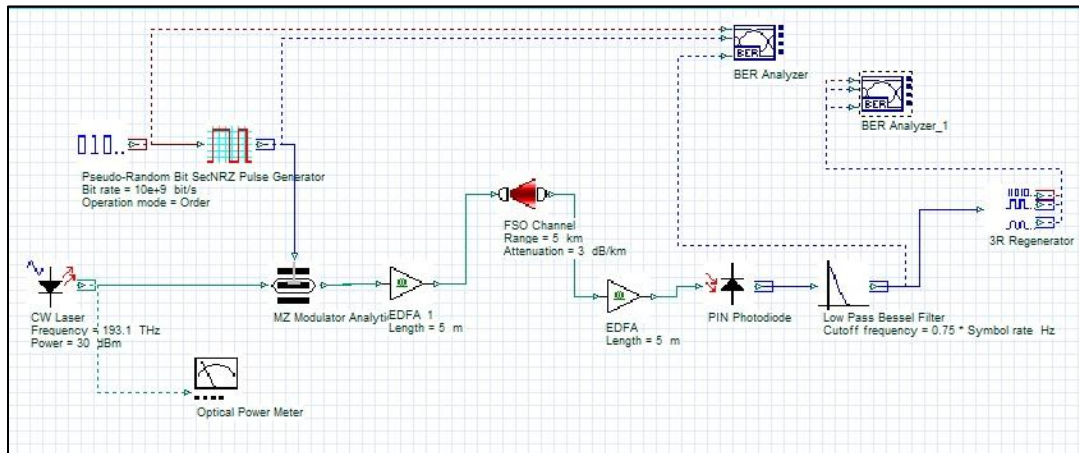


Figure 9 Circuit diagram of FSO

Results:-

Parameters:-

Bit rate = 10 Gbit/s

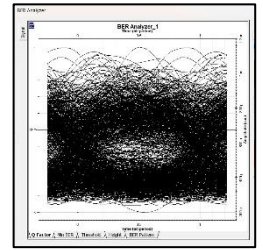
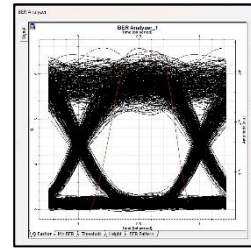
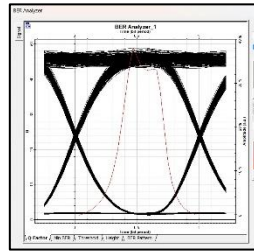
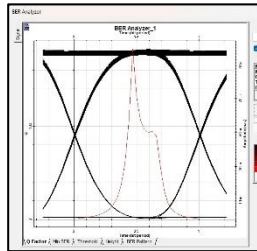
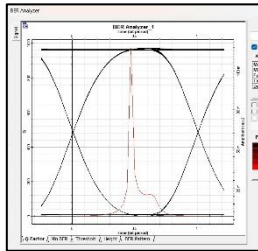
CW frequency = 193.1 THz

CW Laser Power = 25 dBm

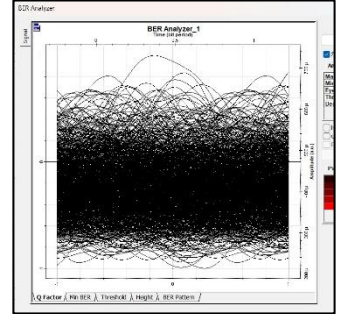
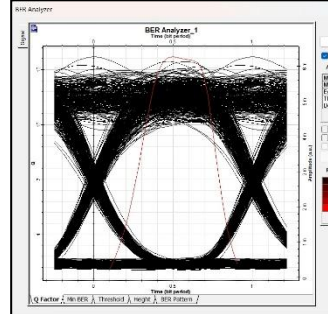
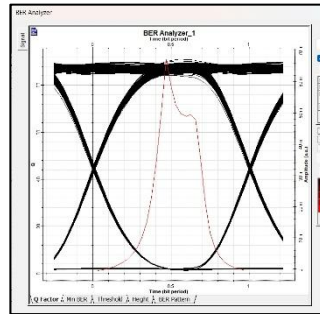
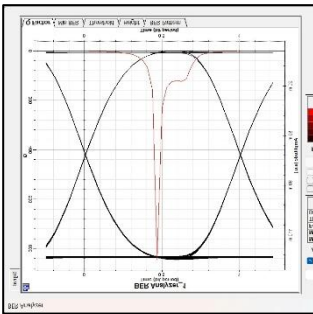
Symbol rate = 10 G Symbol/sec

Extinction ratio of MZ Modulator = 30 dB

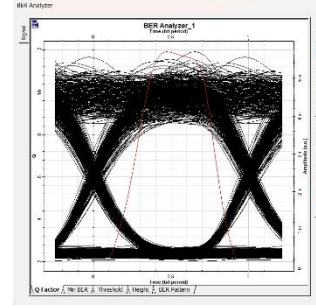
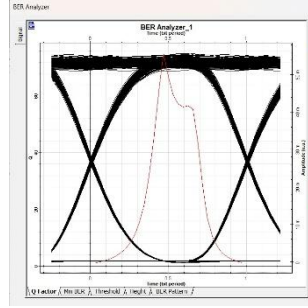
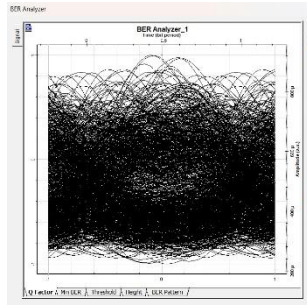
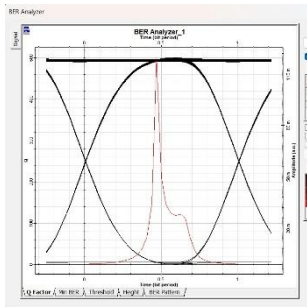
EDFA length = 5 m



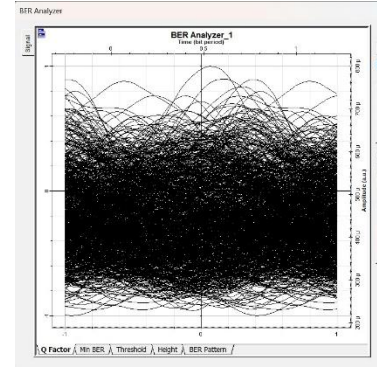
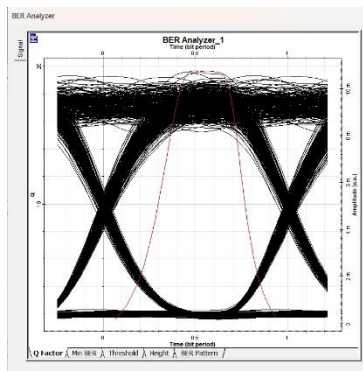
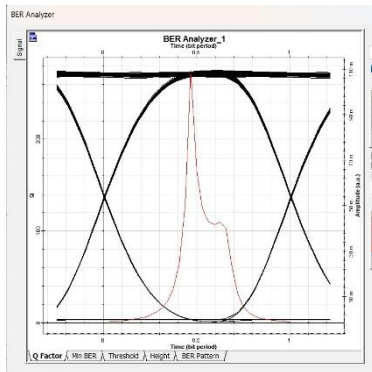
Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Threshold
3 Db/Km	1	553.286	0	0.11758 m	0.01459
	3	90.97	0	0.0642	0.00544
	5	23.53	6.64142e-123	0.01085	0.00166
	8	3.642	0.000121	0.00011	0.000596
	10	0	1	0	0



Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Threshold
5 Db/Km	1	467.44	0	0.1153	0.014
	3	44.11	0	0.02715	0.0032
	5	6.335	1.0266e-10	0.0007	0.00074
	8	0	1	0	0



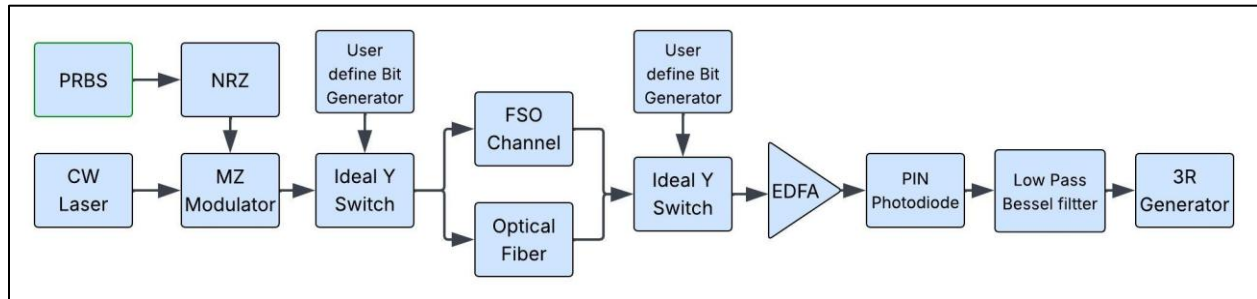
Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Threshold
10 Db/Km	1	246.366	0	0.1054	0.013
	2	36.4742	9.04024e-292	0.02030	0.0022
	3	5.467	1.84156e-08	0.00054	0.00069
	4	0	1	0	0



Attenuation	Range (Km)	Max. Q factor	Min. BER	Eye Height	Threshold
15 Db/Km	1	144.54	0	0.0857	0.01095
	2	9.69825	1.24648e-22	0	
	3	0	1	0	0

3. Hybrid System

Block Diagram:-



- Transmitter:-

The transmitter section consists of a pseudo-random bit sequence (PRBS) generator followed by an NRZ pulse generator, which produces the electrical data signal. This signal is used to drive a Mach-Zehnder Modulator (MZM), where the input optical carrier from a continuous wave (CW) laser operating at 193.1 THz is modulated.

Additionally, user-defined bit sequence generators are incorporated to control the switching mechanism of the hybrid system. These sequences act as control signals for the ideal optical switch, enabling selection between the FSO and optical fiber channels.

- Channel Section:-

The channel section comprises a hybrid transmission link integrating both Free Space Optics (FSO) and optical fiber paths.

The FSO channel is modeled with a transmission range of 3 km and an attenuation factor of approximately 3 dB/km, representing atmospheric losses.

The optical fiber link is also configured with a length of 3 km, providing a low-loss guided transmission medium.

An ideal optical switch is used at the transmitter side to select the active channel based on the control signal. At the receiver side, an ideal optical selector ensures proper routing of the received signal.

To monitor system performance, optical spectrum analyzers and optical power meters are placed along both channels.

- Receiver:-

At the receiver, the incoming optical signal is first amplified using an Erbium-Doped Fiber Amplifier (EDFA) to compensate for transmission losses. The amplified signal is then converted into an electrical signal using a PIN photodiode.

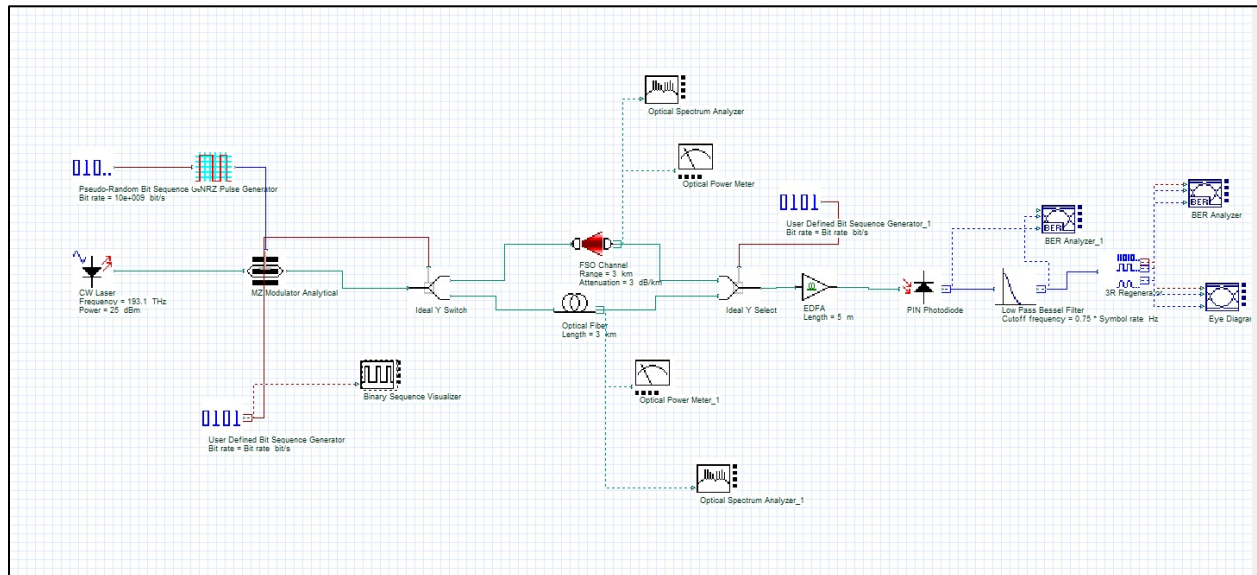
A low-pass Bessel filter is employed to remove high-frequency noise and improve signal quality. The filtered signal is passed through a 3R regenerator (re-amplification, re-shaping, and re-timing) to restore the original signal characteristics.

Finally, system performance is evaluated using a BER analyzer and an eye diagram analyzer, which provide key metrics such as Q-factor, BER, and eye opening.

- Control Signal Design:-

Due to the absence of a feedback mechanism, user-defined bit sequence generators are used at multiple points to manually control the switching and selection process. This approach enables functional validation of the hybrid system, although future implementations can incorporate automatic feedback-based switching for adaptive channel selection.

Simulation



Result:-

Parameter	FSO	Wired System
Q-Factor	73.167	95.996
BER	0	0
Eye Height	0.223	0.0654
Decision Instant	0.531 bit	0.5 bit
Signal Quality	High	Very High
Reliability	High	Very High

1. FSO Channel Performance (Switch = 0)

Under FSO transmission, the optical spectrum exhibits **significant spectral broadening and increased noise components**, primarily due to atmospheric attenuation, scattering, and turbulence effects. The configured attenuation of approximately **3 dB/km** further contributes to signal degradation.

The corresponding eye diagram shows:

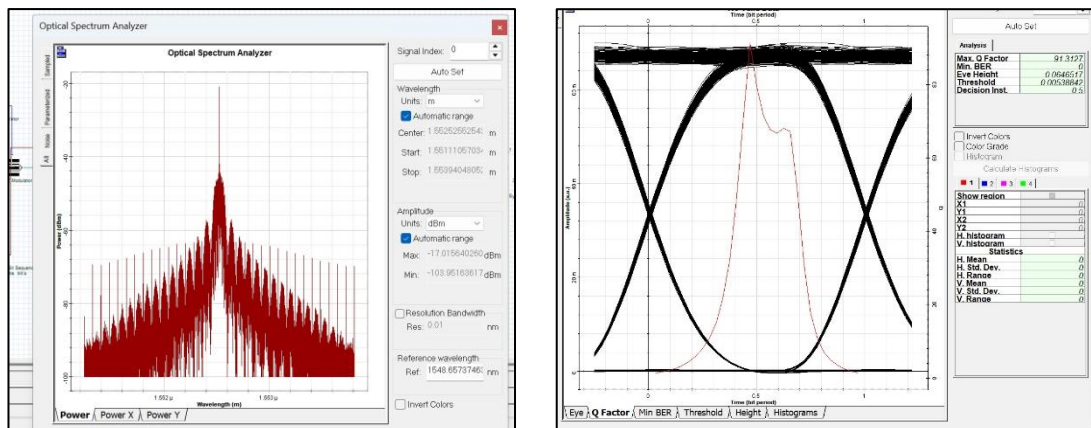
- Reduced eye opening
- Increased timing jitter
- Higher noise margin overlap

The measured **Q-factor (~91)** indicates moderate signal quality, while the **BER is comparatively higher**, confirming performance degradation in the free-space channel.

Inference

The FSO link is highly susceptible to environmental conditions, resulting in:

- Reduced transmission reliability
- Increased error probability
- Degradation in signal integrity



2. Optical Fiber Channel Performance (Switch = 1)

In contrast, the optical fiber channel demonstrates a **well-confined optical spectrum with minimal noise**, indicating low attenuation and negligible external interference.

The eye diagram reveals:

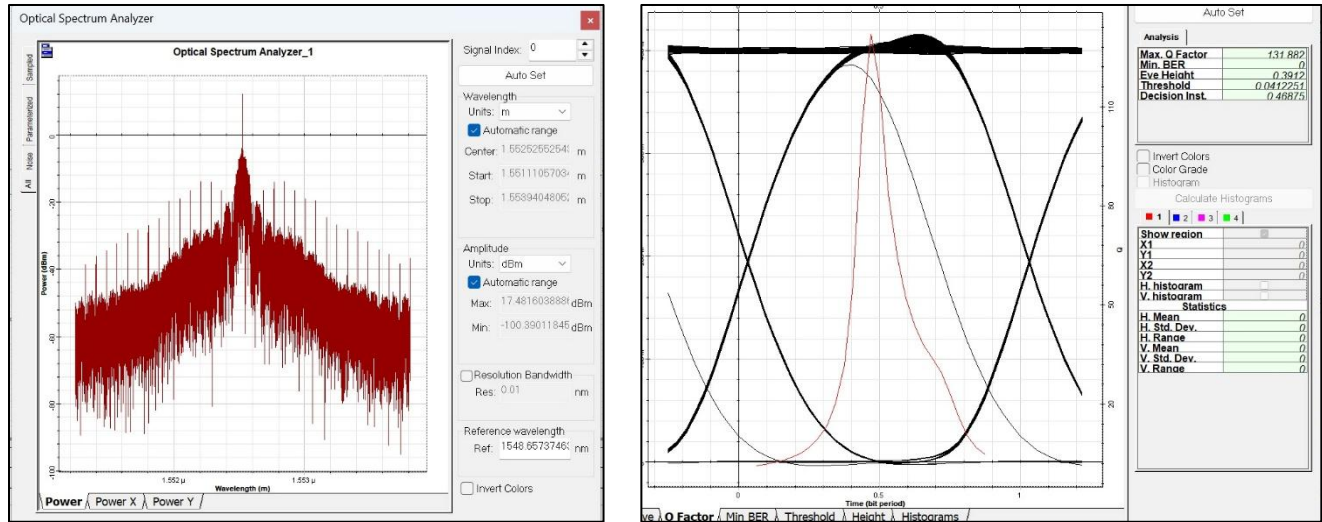
- A wide and well-defined eye opening
- Minimal inter-symbol interference (ISI)
- Negligible timing jitter

The observed **Q-factor (~131)** is significantly higher than that of the FSO channel, and the **BER approaches zero**, indicating highly reliable transmission.

Inference

The fiber optic link ensures:

- High signal fidelity
- Low distortion
- Superior transmission efficiency



Conclusion :-

The hybrid FSO–optical fiber communication system was successfully analyzed using optical spectrum and eye diagram parameters. The performance evaluation demonstrates that the FSO channel is significantly affected by atmospheric conditions, resulting in spectral broadening, reduced eye opening, lower Q-factor, and higher bit error rate (BER). In contrast, the optical fiber channel exhibits a stable and narrow spectrum with a clearly open eye diagram, yielding a higher Q-factor and negligible BER.

A comparative analysis confirms that optical fiber provides superior signal quality, reliability, and transmission efficiency, whereas the FSO link offers flexibility and ease of deployment in scenarios where fiber installation is impractical.

Therefore, the proposed hybrid system effectively combines the advantages of both technologies by enabling adaptive switching between FSO and fiber links, ensuring continuous and reliable communication. This makes the system a robust, scalable, and efficient solution for modern high-speed optical networks.

References:-

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